

Feature Engineering Analytics

	Ticker	ISIN	Name	Region	Country	Sector	Industry
0	NVDA	US67066G1040	NVIDIA Corporation	United States and Canada	US	Information Technology	Semiconductors and Semiconductor Equipment
1	GOOGL	US02079K3059	Alphabet Inc.	United States and Canada	US	Communication Services	Interactive Media and Services
2	AAPL	US0378331005	Apple Inc.	United States and Canada	US	Information Technology	Technology Hardware Storage and Peripherals
3	MSFT	US5949181045	Microsoft Corporation	United States and Canada	US	Information Technology	Software
4	AMZN	US0231351067	Amazon.com Inc.	United States and Canada	US	Consumer Discretionary	Broadline Retail
...	...	...	...	...	...	...	...
6721	SIAME	TN0006590012	Société Industrielle d'Appareillage et de Maté...	Africa / Middle East	TN	Industrials	Electrical Equipment
6722	BERGER	NGBERGER0000	Berger Paints Nigeria Plc	Africa / Middle East	NG	Materials	Chemicals
6723	SOTET	TN0006530018	Société Tunisienne d'Entreprises de Télécommun...	Africa / Middle East	TN	Communication Services	Diversified Telecommunication Services
6724	YPFD	ARP9897X1319	YPF Sociedad Anónima	Latin America and Caribbean	AR	Energy	Oil Gas and Consumable Fuels
6725	HAR	ZAE000015228	Harmony Gold Mining Company Limited	Africa / Middle East	ZA	Materials	Metals and Mining

6726 rows × 252 columns

✓ Dataset loaded: 6,726 stocks, 252 features

Feature Categories Summary:

- Valuation Ratios: 22 features
- Momentum & Technical: 25 features
- Profitability: 15 features
- Quality & Risk: 19 features
- Leverage & Liquidity: 20 features
- Analyst Sentiment: 19 features
- Earnings Quality: 19 features
- Growth Metrics: 14 features
- Dividend Features: 14 features
- Cash Flow: 18 features
- Composite Scores: 5 features

Total engineered features: 190

 Dataset Overview

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Total Stocks: 6,726
Total Columns: 252
Numeric Features: 232
Sectors: 9
Regions: 5
Countries: 92

 Key Statistics:

Avg Market Cap: \$0.00B
Median P/E Ratio: 23.80
Avg Upside Potential: 24.7%

 Coverage Summary by Category:

	mean	min	max
Category			
Analyst Sentiment	99.00	95.20	100.00
Cash Flow	97.90	88.70	100.00
Composite Scores	100.00	100.00	100.00
Dividend Features	79.50	56.70	100.00
Earnings Quality	83.80	52.00	100.00
Growth Metrics	96.00	87.60	100.00
Leverage & Liquidity	94.70	79.90	100.00
Momentum & Technical	99.60	95.10	100.00
Profitability	97.80	94.10	100.00
Quality & Risk	97.80	83.80	100.00
Valuation Ratios	83.00	67.70	97.70

 Valuation Ratios Statistics:

	count	mean	std	min	25%	50%	75%	max	missing_%
p_e_ratio	5302.00	40.10	53.03	0.00	14.90	23.80	41.80	492.50	21.17
p_b_ratio	6525.00	5.34	11.80	0.10	1.40	2.70	5.40	330.60	2.99
ev_ebitda_ratio	6060.00	20.25	34.64	0.20	7.50	11.30	18.90	500.00	9.90
ev_sales_ratio	6570.00	6.19	23.59	0.00	1.10	2.30	4.80	491.80	2.32
peg_ratio	4618.00	1359.05	22440.97	2.60	142.70	283.62	555.78	1385000.00	31.34

Momentum Statistics:

	count	mean	std	min	25%	50%	75%	max
price_momentum_1m	6722.00	7.37	23.59	-74.13	-0.71	4.47	11.94	1505.10
price_momentum_3m	6702.00	8.91	31.63	-74.77	-5.15	4.50	17.84	1484.91
price_momentum_6m	6672.00	22.15	58.54	-98.02	-6.39	9.46	33.75	1455.56
price_momentum_1y	6600.00	36.68	87.27	-97.74	-7.05	15.58	51.14	1918.00
range_52w_position	6726.00	0.63	0.30	0.00	0.39	0.70	0.90	1.00

Quality Scores by Sector:

	piotroski_f_score	quality_momentum_score	dilution_score	accounting
_quality_score	earnings_quality_score			
Sector				
Communication Services		5.40	46.10	50.90
79.40	80.30			
Consumer Discretionary		5.00	48.50	51.80
83.80	83.20			
Consumer Staples		5.20	47.10	51.10
83.30	84.10			
Energy		4.90	51.20	48.00
88.60	78.60			
Health Care		4.70	54.30	45.50
85.30	78.00			
Industrials		5.00	51.00	49.80
84.70	83.90			
Information Technology		5.00	55.20	48.00
83.80	75.20			
Materials		5.00	53.60	45.80
84.90	83.90			
Utilities		5.00	42.40	48.50
88.80	85.80			

Sentiment Statistics:

	count	mean	std	min	25%	50%	75%	max
analyst_bullish_pct	6633.00	66.98	30.81	0.00	48.00	72.73	100.00	100.00
analyst_bearish_pct	6633.00	6.43	14.68	0.00	0.00	0.00	7.14	100.00
upside_potential	6725.00	24.66	377.77	-100.00	1.86	13.73	30.43	30574.85
analyst_rating_normalized	6726.00	75.53	21.39	-25.00	66.00	79.50	90.19	100.00
eps_revision_momentum	6726.00	0.08	4.75	-0.85	-0.02	0.00	0.02	386.70

Regional Averages:

	p_e_ratio	roe	debt_to_equity	upside_potential	dividend_yield
quality_momentum_score					
Region					
Africa / Middle East	30.01	0.18	0.77	25.18	0.04
48.55					
Asia / Pacific	49.74	0.12	0.45	9.99	0.02
54.10					
Europe	32.25	0.10	0.51	41.36	0.04
47.79					
Latin America and Caribbean	32.68	0.40	1.48	38.09	0.05
45.93					
United States and Canada	41.12	0.02	0.80	19.93	0.03
52.74					

Cash Flow Quality Statistics:

	count	mean	std	min	25%	50%	75%	max
cfo_to_net_income	6720.00	0.94	33.03	-1338.00	0.58	1.38	2.31	1042.81
fcf_margin	6614.00	-1.09	37.82	-2246.11	0.00	0.06	0.14	284.66
fcf_positive_ratio	6726.00	0.63	0.36	0.00	0.40	0.80	1.00	1.00
cash_flow_quality_score	6726.00	61.74	35.26	0.00	25.00	75.00	100.00	100.00
self_funding_ratio	6497.00	4.29	56.48	-1501.53	0.71	1.60	3.29	2133.27

Top 10 Stocks by Quality Momentum Score:

	Ticker	Name	Sector	quality_momentum_score
14	MU	Micron Technology Inc.	Information Technology	90.00
29	LRCX	Lam Research Corporation	Information Technology	90.00
308	COHR	Coherent Corp.	Information Technology	90.00
472	HL	Hecla Mining Company	Materials	90.00
593	GMED	Globus Medical Inc.	Health Care	90.00
649	ONTO	Onto Innovation Inc.	Information Technology	90.00
753	7769	Hon. Precision Inc.	Information Technology	90.00
1813	NUTX	Nutex Health Inc.	Health Care	90.00
2317	TPRO	Technoprobe S.p.A.	Information Technology	90.00
2727	COPN	Cosmo Pharmaceuticals N.V.	Health Care	90.00

Top 10 Stocks by Upside Potential (with analyst coverage):

	Ticker	Name	Sector	upside_potential
4050	PRO	ProService Building Services Marketplace Plc	Industrials	30574.85
6656	AMBP3	Ambipar Participações e Empreendimentos S.A.	Industrials	4185.71
2700	ALSEN	Sensorion SA	Health Care	652.51
3985	MRSHL	Marshall Boya Ve Vernik Sanayii A.S.	Materials	603.82
3509	FARN	Faron Pharmaceuticals Oy	Health Care	477.53
3694	SAVE	Savannah Energy PLC	Energy	411.23
3605	PANR	Pantheon Resources Plc	Energy	400.00
3977	ZPHR	Zephyr Energy plc	Energy	366.67
3768	NVM	Novem Group S.A.	Consumer Discretionary	364.63
3913	TPG0	The Platform Group AG	Consumer Discretionary	325.08

Top 10 Stocks by Piotroski F-Score:

	Ticker	Name	Sector	piotroski_f_score
79	NEM	Newmont Corporation	Materials	9
250	K	Kinross Gold Corporation	Materials	9
351	STE	STERIS plc	Health Care	9
370	CW	Curtiss-Wright Corporation	Industrials	9
422	INCY	Incyte Corporation	Health Care	9
575	CHWY	Chewy Inc.	Consumer Discretionary	9
621	EXEL	Exelixis Inc.	Health Care	9
766	AWI	Armstrong World Industries Inc.	Industrials	9
799	PAYC	Paycom Software Inc.	Industrials	9
904	DAR	Darling Ingredients Inc.	Consumer Staples	9

 Feature Statistics Summary (first 20 features):

	category	feature	count	mean	median	std	min	q25	q75	
0	Valuation Ratios	p_e_ratio	5302	40.10	23.80	53.03	0.00	14.90	41.80	
1	Valuation Ratios	p_b_ratio	6525	5.34	2.70	11.80	0.10	1.40	5.40	
2	Valuation Ratios	ev_ebitda_ratio	6060	20.25	11.30	34.64	0.20	7.50	18.90	
3	Valuation Ratios	ev_sales_ratio	6570	6.19	2.30	23.59	0.00	1.10	4.80	
4	Valuation Ratios	dividend_yield	4636	0.03	0.02	0.06	0.00	0.01	0.04	
5	Valuation Ratios	peg_ratio	4618	1359.05	283.62	22440.97	2.60	142.70	555.78	1385
6	Valuation Ratios	ev_sales_trend_1y	6272	0.47	0.07	3.41	-1.00	-0.17	0.44	
7	Valuation Ratios	ev_ebitda_momentum	5762	1.17	0.05	26.58	-1.00	-0.20	0.44	1
8	Valuation Ratios	p_e_momentum_yoy	5053	0.81	0.16	4.18	-1.00	-0.20	0.78	
9	Valuation Ratios	p_e_momentum_qoq	5259	0.35	0.15	1.46	-1.00	-0.05	0.45	
10	Valuation Ratios	ev_sales_vs_3y_avg	6141	0.17	0.05	0.64	-1.00	-0.12	0.31	
11	Valuation Ratios	ev_ebitda_vs_3y_avg	5626	0.14	0.03	0.58	-0.99	-0.15	0.28	
12	Valuation Ratios	p_e_vs_3y_avg	4923	0.29	0.11	0.93	-1.00	-0.16	0.45	
13	Valuation Ratios	ev_sales_forward_discount	6416	-0.03	-0.06	1.38	-0.98	-0.14	0.00	
14	Valuation Ratios	ev_ebitda_forward_discount	5873	-0.08	-0.12	1.54	-0.98	-0.26	-0.01	
15	Valuation Ratios	p_e_forward_discount	5021	-0.05	-0.17	2.08	-1.00	-0.36	-0.02	
16	Valuation Ratios	p_b_vs_5y_avg	5516	1.14	1.00	0.79	0.03	0.73	1.32	
17	Valuation Ratios	ev_sales_qoq_1q	6499	0.39	0.07	8.07	-1.00	-0.07	0.29	
18	Valuation Ratios	p_e_vs_5y_avg	4554	0.24	0.02	1.10	-1.00	-0.27	0.41	

	category	feature	count	mean	median	std	min	q25	q75
19	Valuation Ratios	valuation_compression	4866	0.20	0.09	0.60	-0.87	-0.13	0.36

✓ Saved statistics to outputs/feature_statistics_summary.csv

 Available Features for Exploration:

1. accounting_quality_score
2. accumulated_deficit_flag
3. acquisition_intensity
4. adequate_cash_buffer
5. altman_z_score
6. altman_z_trend
7. analyst_bearish_pct
8. analyst_bullish_pct
9. analyst_coverage_change_1m
10. analyst_coverage_change_3m
11. analyst_coverage_quality
12. analyst_rating_normalized
13. asset_quality_score
14. asset_turnover
15. balance_sheet_strength
16. beta_momentum
17. breakout_signal
18. buyback_yield
19. capex_vs_5y_avg
20. cash_burn_rate
21. cash_change_qoq
22. cash_flow_quality_score
23. cash_ratio
24. cash_runway_months
25. cash_to_assets_pct
26. cash_vs_5y_avg
27. cfo_growth_yoy
28. cfo_positive_quarters
29. cfo_quarterly_trend
30. cfo_to_net_income
- ... and 160 more